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RESEARCH PAPER

Impaired hydration status in acutely admitted older patients: prevalence and impact on mortality

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Abstract

Background: impaired hydration is common in the older people, however studies of its effects on outcome in the acute setting are limited.
Objectives: to assess (i) the prevalence of impaired hydration, (ii) its relationship with laboratory markers of altered hydration and with (iii) short- and long-term mortality.
Design: retrospective cohort study.
Setting: University Hospital-Internal Medicine Department.
Subjects: a total of 5,113 older patients consecutively acutely admitted from October 2015 to July 2016.
Methods: according to calculated serum osmolarity at admission hydration status was stratified in: low osmolarity (<275 mmol/L), euhydration (275–295 mmol/L), impending (296–300 mmol/L) and current dehydration (>300 mmol/L). Relationships with serum sodium, potassium, glucose, urea, estimated glomerular filtration rate (eGFR), haematocrit, urea/creatinine ratio (Urea/Cr) and urine specific gravity (USG) were determined. Charlson Comorbidity Index, Modified Early Warning Score, Glasgow Prognostic Score, Norton score and Nutritional Risk Screening-2002 were calculated.
Results: current and impending dehydration, euhydration and low-osmolarity were detected in 51.7, 17.1, 28.5 and 2.7% of the patients, respectively. Osmolarity correlated with urea ($r = 0.846$). Associations with serum sodium, creatinine, eGFR and urea/Cr were low but significant, being negligible that with USG and haematocrit. Serum sodium and urea increased in the transition from low- to high-osmolarity ($P < 0.001$ in all pairwise comparisons). In multivariate modelling current dehydration, functional dependence, clinical instability and high nutritional risk were associated ($P < 0.001$) with reduced short- and long-term survival.
Conclusions: impaired hydration is common in older people acutely admitted to medical care and is associated with poor outcome. Early assessment of calculated serum osmolarity is mandatory to target dehydration and hypoosmolar disorders.

Keywords: hydration, older people, mortality, hospital, serum osmolarity

Key Points

- Prevalence of impaired hydration in acute older patients is high.
- Dehydration is associated with increased short- and long-term mortality.
- Hypoosmolarity is likely to increase mortality at early and late time points.

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Introduction

Acute changes of hydration status are common in older people and are associated with adverse outcomes acutely and chronically [1]. The assessment of hydration status is hampered by the low accuracy of clinical signs and of instrumental techniques used to evaluate total body water. In contrast, directly measured serum osmolality is considered one of the most accurate markers of hyperosmolar dehydration [2]. Given that directly measured serum osmolality is rarely assessed in routine clinical practice, serum osmolarity can be pragmatically calculated from routinely measured serum biochemistry by applying validated equations [3]. Since the difference between directly measured osmolality and calculated osmolarity is negligible and in most cases clinically irrelevant [2], calculated serum osmolarity is widely accepted as reference standard for hydration status in older patients, with standardised cut offs to diagnose current and impending dehydration [4].

Dehydration is characterised by a deficit in total body water. Hyperosmolar dehydration (also defined extracellular dehydration) can occur either because of inadequate fluid intake or of excess renal losses, resulting in increased serum osmolarity [5]. Hyperosmolar dehydration is a predictor of in-hospital and post-discharge mortality in older patients [2,6]. The percentage of older people presenting at hospital admission with hyperosmolar dehydration is high [2,6,7] and unequivocally associated with poor clinical outcomes [8]. For this reason, actions should be undertaken in the presence of impending dehydration [5,9], because at this stage preventive measures are still possible to avoid complications. So far, data on prevalence and outcome of impending dehydration in older patients acutely admitted to medical care is limited [10]. In addition, very few studies have analysed the prevalence and impact of hypoosmolar disorders. Finally, many measures derived from laboratory tests (e.g. urea/creatinine ratio [Urea/Cr], estimated glomerular filtration rate [eGFR], haematocrit, urine specific gravity [USG]) are used in the clinical practice as surrogate markers of hydration. Their correlation with calculated serum osmolarity has not been addressed before. Therefore, this study aimed to investigate the prevalence of altered hydration status in acutely hospitalised older people and its impact on short-term and long-term mortality. Moreover, the relationships between serum osmolarity and several laboratories and clinical parameters were explored.

Participants and Methods

Study design, setting and population

This was a retrospective observational study conducted in the 650-bed University Hospital of Trieste, Italy. All consecutive acute patients aged >65 who were admitted from the emergency room to the Internal Medicine Department between 15 October 2015 and 31 July 2016 were considered for eligibility. Patients with elective hospital admission or with incomplete data to calculate serum osmolarity were excluded.

Ethics

This research is part of the PROPOSING IDEAS (PROgnostic factors of Poor hospital Outcome based on clinical Severity, nutrItional and NursinG indexes IDEntified on AdmiSsion in elderly/frail patients) study, approved by the Regional Bioethics Committee of Friuli Venezia Giulia, Italy (protocol number: 28,217; date of approval: 31 July 2018). The research was conducted according to the Declaration of Helsinki. At hospital admission, all enrolled patients authorised the use of their clinical data for study purposes.

Study variables

Clinical, functional and laboratory data assessed on admission were collected from the hospital electronic archive and from the electronic nursing assessment register. Serum osmolarity was calculated based on the equation by Krahn and Khajuria [11], validated in very different populations of older adults [4]:

$$1.86 \times \left(Na^{+} + K^{+}\right) + 1.15 \times \text{glucose} + \text{urea} + 14$$

All laboratory data in the equation were converted in mmol/L. Based on calculated serum osmolarity patients were categorised in the following groups: (i) euhydration (275–295 mmol/L), (ii) impending dehydration (296–300 mmol/L), (iii) current dehydration (>300 mmol/L) and (iv) hypo-osmolarity (<275 mmol/L).

The eGFR was calculated using the formula by Levey *et al*. [12]. Renal function was defined as normal-to-mildly decreased (eGFR ≥60) or moderately-to-severely decreased (eGFR <60 ml/min/1.73 m^2), being eGFR <60 ml/min a marker of increased risk of mortality in the oldest old [13].

Haematocrit, USG and Urea/Cr were assessed as widely used markers of dehydration, despite previously described limits in discriminating hydration status among older people [7,10,14,15]. Urea/Cr >20 [16] and haematocrit >50% (laboratory threshold) were identified as thresholds for possible dehydration.

The Glasgow Prognostic Score (GPS), an inflammation-based prognostic marker based on C-reactive protein (CRP) and albumin levels [17], was calculated and the following categories were identified: 0 (good prognosis—CRP ≤10 mg/L and albumin ≥3.5 g/dL), 1 (intermediate prognosis—CRP >10 mg/L and albumin ≥3.5 g/dL, or CRP ≤10 mg/L and albumin <3.5 g/dL) and 2 (poor prognosis—CRP >10 mg/L and albumin <3.5 g/dL).

Anaemia is an independent risk factor for in-hospital mortality, particularly when associated with renal dysfunction [18]. Patients presenting with haemoglobin <8 g/dL were considered at high risk.

Nutritional Risk Screening-2002 (NRS-2002) was used to detect the risk of malnutrition. NRS-2002 ≥3 has been

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previously identified as an independent predictor of early and late mortality in acute hospitalised patients [19].

The Modified Early Warning Score (MEWS, range 0–13) [20] was employed to assess patient criticality on the basis of physiological vital signs. Low (≤2), intermediate (3–4) and high (≥5) risk categories were identified.

The Norton scale [21], a risk assessment tool (range 5–20) considering patient's overall functional status and associated with long-term mortality in older people [22], was used, identifying the following risk categories: low (≥19), intermediate (14–18), high (10–13) and very high (≤9).

The age-adjusted Charlson Comorbidity Index (CCI) was calculated based on patients' medical history [23]. The risk of death was considered very high for CCI ≥5.

The date of patient's death, irrespective of the cause, was used to compute early (up to 30 days) and late (3-12 months) mortality rates after hospital admission.

Statistical analysis

Data distribution was evaluated using the Kolmogorov–Smirnov test. Continuous variables were described as median and interquartile range (IQR). Differences between subgroups were analysed via nonparametric Kruskal–Wallis test adjusted by the Bonferroni correction for multiple pairwise comparisons. Correlation between continuous variables was performed with Pearson's coefficient (r). Correlation strengths were defined as follows: 0–0.30, negligible; 0.30–0.50, low; 0.50–0.70, moderate; 0.70–0.90, high and 0.90–1, very high [24].

Multivariate Cox regression models were run to estimate patient's risk of death at different intervals according to calculated serum osmolarity, adjusted for confounders potentially acting as surrogate marker of dehydration or known to potentially impact on mortality. Age *per se* was not considered as a predictor being already computed in both CCI and NRS-2002. USG was not deemed because of the relevant number of missing data. Stepwise method was adopted (P values: entry 0.01; removal 0.05) to identify the best subset of predictors for each considered mortality interval. Observations were right-censored until the corresponding number of days (e.g. 7, 15, 30, . . . , 365) from admission.

The assumption of Cox models was assessed by comparing the log–log transformation versus the log of survival time of the Kaplan–Meier curves for the osmolarity groups. Since the curves were approximately parallel and did not intersect with time, a constant over-time proportional hazard was assumed.

An alpha level of $P < 0.01$ was set to determine statistical significance to reduce type I error rate. Statistical analyses were performed using SPSS for Windows, version 23.0 (IBM Corp., Armonk, NY, USA).

Results

During the study period, 5,113 patients aged ≥65 were assessed for inclusion. Five-hundred (9.8%) patients were excluded due to incomplete laboratory data for osmolarity computing, resulting in the inclusion of 4,613 patients. Table 1 shows the characteristics of the study population. The prevalence of current and impending dehydration, euhydration and hypoosmolar conditions were 51.7, 17.1, 28.5 and 2.7%, respectively.

Table 1. Baseline characteristics of enrolled population at hospital admission

Variable	Data
Age (years)	84.0; 78.0–89.0
Sex (female)	58.0% (n = 2,675)
CCI[a]	7.0; 5.0–8.0
CCI ≥ 5	85.3% (n = 3,934)
GPS[b]	
Score 0	24.8% (n = 1,093)
Score 1	29.6% (n = 1,366)
Score 2	42.3% (n = 1,953)
MEWS[c]	3.0; 2.0–5.0
MEWS ≥ 5	29.5% (n = 1,357)
Nutritional risk score-2002	2.0; 2.0–3.0
Nutritional risk score-2002 ≥ 3	39.4% (n = 1,819)
Norton scale[d]	15.0; 11.0–18.0
Norton scale ≤ 9	20.3% (n = 681)
Serum sodium (mEq/L)	136.0; 133.0–139.0
Serum potassium (mEq/L)	4.0 (3.7–4.4)
Blood glucose (mg/dL)	120.0; 102.0–149.0
Blood urea (mg/dL)	47.0; 35.0–68.0
Calculated osmolarity (mmol/L)	301.0; 293.7–310.6
<275 mmol/L	2.7% (n = 125)
275–295 mmol/L	28.5% (n = 1,314)
296–300 mmol/L	17.1% (n = 789)
>300 mmol/L	51.7% (n = 2,385)
Urea to creatinine ratio	46.6; 37.0–59.1
Urea to creatinine ratio > 20	98.6% (n = 4,547)
eGFR (ml/min/1.73 m^2)	64.0; 45.1–84.6
eGFR < 60 ml/min/1.73 m^2	44.6% (n = 2,059)
Haemoglobin (g/dL)	11.9; 10.5–13.2
Haemoglobin < 8 g/dL	3.7% (n = 169)
Haematocrit (%)	37.7; 33.3–41.6
Urine specific gravity[e]	1015.0; 1011.0–1020.0

Data are reported as 'median; IQR' or 'percentage (frequency)'. [a]n = 4,392; [b]4,412; [c]4,598; [d]3,357; [e]n = 2,587

Significant correlations between serum osmolarity and biochemistry were observed (Figure 1) with different degrees of strength (urea: $r = 0.846$; serum sodium: $r = 0.634$; creatinine: $r = 0.549$; eGFR: $r = -498$; Urea/Cr: $r = 0.442$). A negligible correlation was demonstrated with the other explored laboratory test variables (glucose: $r = 0.251$; potassium: $r = 0.166$; USG: $r = 0.071$; haematocrit: $r = -0.051$).

In pairwise comparisons, serum sodium and urea showed a progressive ($P < 0.001$) increase concordantly with raising osmolarity categories (Figure 2, a and d). Among other potential markers of hydration only eGFR and Urea/Cr partially showed a similar trend, which, however, did not reach statistical significance in the comparison with the <275 mmol/L subgroup (Figure 2, f and g). No comorbidity, functional, severity-related, and nutritional scores showed a similar progression, although patients with current dehydration had a significantly higher CCI (Figure 2, i–n).

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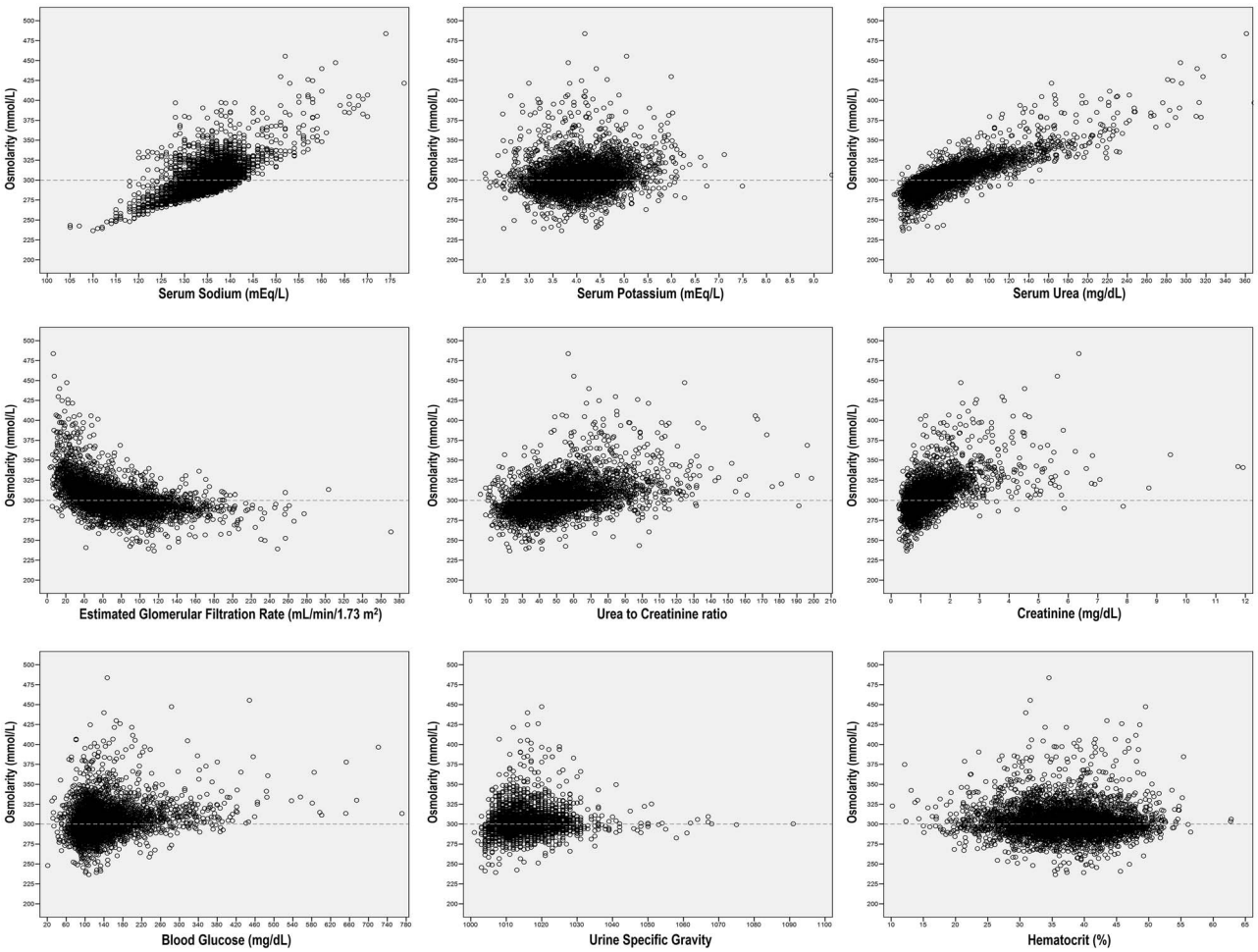

Figure 1. Scatterplots representing the correlation analyses between serum osmolarity and laboratory test variables considered in the study.

Mortality data within 12-months from hospital admission were available in 99.5% of cases (n = 4,589). The cumulative mortality rate was 5.7, 9.8, 14.5, 23.6, 31.1, and 39.1% at 7, 15, 30, 90, 180, and 365 days, respectively. According to the multivariate Cox model adjusted for sex, MEWS, Norton index, NRS-2002, GPS, CCI, haemoglobin, eGFR, haematocrit and Urea/Cr; patients with current dehydration showed a statistically significant increased risk of death at every considered time interval (hazard ratio [HR] ranging from 1.266 to 3.065), whereas neither impending dehydration nor hypo-osmolarity showed to be predictive of mortality (Table 2, Figure 3). Bivariate comparisons between adjustment covariates and mortality are available as Supplementary data.

Discussion

This study is one of the very few which has examined the relationship between the whole spectrum of calculated serum osmolarity as a marker of hydration status and outcome in acute older patients. The main result of the present study is that current dehydration—but not impending dehydration—was an independent predictor of mortality at all considered time points. It is therefore important to early identify this condition in order to set up appropriate treatments at this point to prevent further complications.

Our results showed a high prevalence of dehydration at admission, in line with previous reports in acute settings [2,6]. Pre-existing dehydration is notably linked to age-related pathophysiological changes (e.g. increased thirst threshold, reduced kidney functionality) [1] and may reflect the quality of care provided in community settings [25], sometimes not attentive enough in controlling these pathophysiological aspects to maintain a good hydration state. It is important to notice that serum osmolality is a good marker for assessing acute changes in hydration status, whereas it is not representative of hydration chronically—as it constantly changes [1].

Among considered biochemistry, only serum sodium and urea showed a significant progressive increase in the transition from a lower to a higher osmolarity category, with only urea highly correlated to calculated serum osmolarity. This finding is of particular interest when considering the

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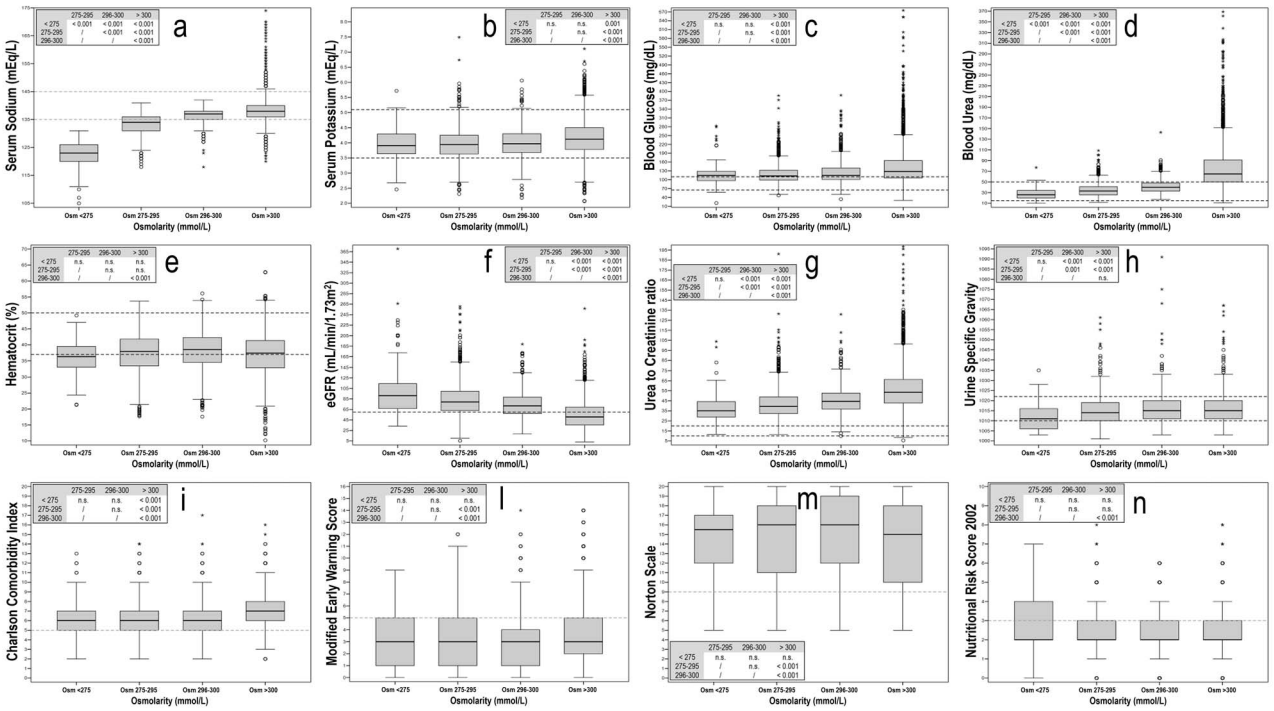

Figure 2. Differences among osmolarity subgroups for laboratory data values considered as determinants for osmolarity calculation (a–d) or as surrogate markers of hydration (e–h), and for comorbidity, functional, acuity and nutritional indexes (i–n). Black horizontal line inside the boxes: median. Boxes height: IQR. Whiskers: 1.5 × IQR. Small circles: outliers. Stars: extreme values. Dashed lines: normal ranges/thresholds. Values in tables embedded in each graph report statistical differences between subgroups (see methods); ns: non-statistically significant.

different impact of the osmolarity equation's components on cell hydration. Although increased urea and sodium can both contribute to serum osmolarity, their impact on cellular hydration is different. Indeed, sodium is an effective osmole: being unable to cross the cell membrane, changes in its serum concentration induce transmembrane water flow, thus increasing or decreasing cell volume. Alterations in cellular hydration may thus critically determine either cell swelling or shrinkage, compromising structural and functional cell integrity up to apoptotic cell death [26]. Conversely, urea is an ineffective osmole since—being able to cross the cell membrane—it produces no transmembrane water flow and, thus, it does not affect cell volume [27]. Consequently, in hyperosmolar patients it is important to identify which of the molecules included in the formula are responsible for the increase in calculated serum osmolarity, due to the different clinical impact that the osmotically active or inactive molecules have on cellular tonicity and, consequently, the different treatment to provide.

Similar considerations also apply to Urea/Cr ratio and eGFR, whose values—despite a low correlation with calculated serum osmolarity (Figure 1)—suggested an almost constant and progressive reduction in renal function as osmolarity increases (Figure 2). Although Urea/Cr ratio as a marker of the hydration status presents the limit of being affected by renal function [14], our findings confirm the previously reported association between dehydration and kidney failure in acute patients [2]. In our study USG and haematocrit showed a negligible correlation with calculated serum osmolarity. These findings confirm that neither USG nor haematocrit should be used to identify impending and overt dehydration in older patients in accordance with current recommendations [10].

A further novelty of our study is having documented for the first time the prevalence of older patients with low serum osmolarity at hospital admission. This subgroup of patients deserves special considerations regarding the implication of this condition in daily clinical practice. Hypoosmolar disorders include patients characterised by concomitant reduced osmolarity and hyponatraemia associated to very different conditions. Indeed, in patients with hypoosmolar-hyponatraemia the extracellular fluid volume can either be: decreased—because of renal (e.g. diuretic agents, osmotic diuresis, adrenal insufficiency) or extra renal (e.g. diarrhoea, vomiting, excessive sweating, fluid sequestration in 'third space', bowel obstruction) sodium loss; increased—in case of congestive heart failure, acute/chronic renal failure, nephrotic syndrome, cirrhosis or excessive water intake; or fundamentally normal—in the presence of endocrinopathies such as antidiuresis syndrome, glucocorticoid deficiency or hypothyroidism [28]. However, hyponatraemia can be associated either to low, normal or even increased

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Table 2. Results of stepwise Cox regression of mortality at different time interval from hospital admission on study variables

Variable	7-days mortality HR; 95% CI	15-days mortality HR; 95% CI	30-days mortality HR; 95% CI	90-days mortality HR; 95% CI	180-days mortality HR; 95% CI	365-days mortality HR; 95% CI
Euhydration	1.000	1.000	1.000	1.000	1.000	1.000
Hypoosmolar condition	2.276; 0.491–10.549*	0.957; 0.294–3.115*	1.141; 0.524–2.486*	0.831; 0.462–1.496*	1.203; 0.801–1.805*	1.148; 0.797–1.653*
Impending dehydration	0.842; 0.259–2.736*	0.835; 0.456–1.530*	1.024; 0.681–1.540*	0.832; 0.622–1.113*	0.904; 0.719–1.135*	0.907; 0.747–1.101*
Current dehydration	3.035; 1.499–6.146**	1.952; 1.340–2.843***	1.735; 1.316–2.287***	1.532; 1.267–1.853***	1.308; 1.115–1.533**	1.266; 1.103–1.453**
MEWS ≥ 5	4.298; 2.349–7.716***	2.426; 1.749–3.364***	1.800; 1.417–2.286***	1.443; 1.215–1.715***	1.380; 1.193–1.596***	1.313; 1.155–1.492***
Norton index ≤ 13	3.390; 2.012–5.713***	2.775; 2–003-3.845***	2.341; 1.835–2.987***	2.192; 1.834–2.620***	2.197; 1.880–2.566***	2.042; 1.776–2.348***
NRS-2002 ≥ 3	2.638; 1.579–5.713***	1.880; 1.387–2.550***	1.954; 1.553–2.459***	1.910; 1.621–2.250***	1.711; 1.491–1.962***	1.547; 1.373–1.744***
GPS = 2	[a]	2.572; 1.792–3.690***	3.702; 2.783–4.926***	3.127; 2.587–3.781***	2.569; 2.206–2.993***	2.205; 1.940–2.507***
Charlson index ≥ 5	[a]	[a]	[a]	[a]	1.708; 1.244–2.346**	1.889; 1.433–2.490***
Sex (female)	[a]	[a]	[a]	[a]	0.755; 0.658–0.866***	0.785; 0.696–0.885***
Haemoglobin < 8 g/dL	[a]	[a]	[a]	[a]	[a]	[a]
eGFR < 60 ml/min/1.73 m^2	[a]	[a]	[a]	[a]	[a]	[a]
Haematocrit > 50%	[a]	[a]	[a]	[a]	[a]	[a]
Urea:Creatinine ratio > 20	[a]	[a]	[a]	[a]	[a]	[a]
Log-likelihood	*1012.988****	*2650.150****	*4633.293****	*8930.869****	*12733.585****	*16836.514****

CI: confidence interval. [a]variable excluded from the final model. *not statistically significant ($P > 0.2$); **$P = 0.001$; ***$P < 0.001$.

serum osmolarity, with very different clinical consequences. When associated to low serum osmolarity, hyponatraemia leads to cell overhydration that can induce brain swelling and dysfunction. Despite this dramatic condition, most patients could be pauci- or asymptomatic, especially when the disorder has developed slowly [28]. Therefore, in the presence of hyponatraemia, an early assessment of serum osmolarity is crucial.

Whatever the aetiology, hypoosmolar conditions have a direct impact on increased morbidity, mortality and health-care costs [29], even in case of moderate hyponatraemia with asymptomatic/pauci-symptomatic patients [30]. In our population, hypoosmolar patients showed a trend towards reduced cumulative survival both at very early and late time points after hospital admission. We believe that the failure to achieve the statistical significance could be due to the low prevalence (2.7%) of this condition. What seems to emerge clearly from our results is that, in any case, this osmolarity threshold may mark upon hospital admission a subgroup of patients at high risk of death. These findings warrant further investigation by increasing sample size in order to confirm the trend and to study the relationships with admission diagnosis and pre-existing diseases.

In very short-term adjusted survival analysis, dehydration, clinical instability, high risk of malnutrition and functional dependence were prognostic of poor survival, therefore a predictive model for mortality including these factors could contribute to early identify patients at increased risk of in-hospital death. At variance with previous studies [6], comorbidities were not statistically significant in the model for short-term mortality, but they significantly contributed to mortality at later time-points. A similar trend was demonstrated for gender (Table 2).

This study has strengths and limitations. A point of strength is having documented—for the first time—the prevalence on hospital admission of four different hydration classes in a large population of older adults and having reported the independent impact of such conditions on short- and long-term mortality after adjusting for relevant clinical and functional variables. Limitations include exclusion of nearly 10% of the eligible patients due to incomplete data to calculate osmolarity, a certain number of missing data for some study variables and the observational retrospective design, which prevented more accurate considerations on some variables of interest (e.g. medical diagnoses). All above factors must be taken into account before generalising the present study results.

In conclusion, acute older patients presenting with dehydration are at increased risk of mortality; those with a hypoosmolar condition are likely to be at increased risk as well, despite the non-conclusive findings of the present study. Therefore, a very early assessment of serum osmolarity upon hospital admission is strongly recommended for risk stratification, to be considered together with other

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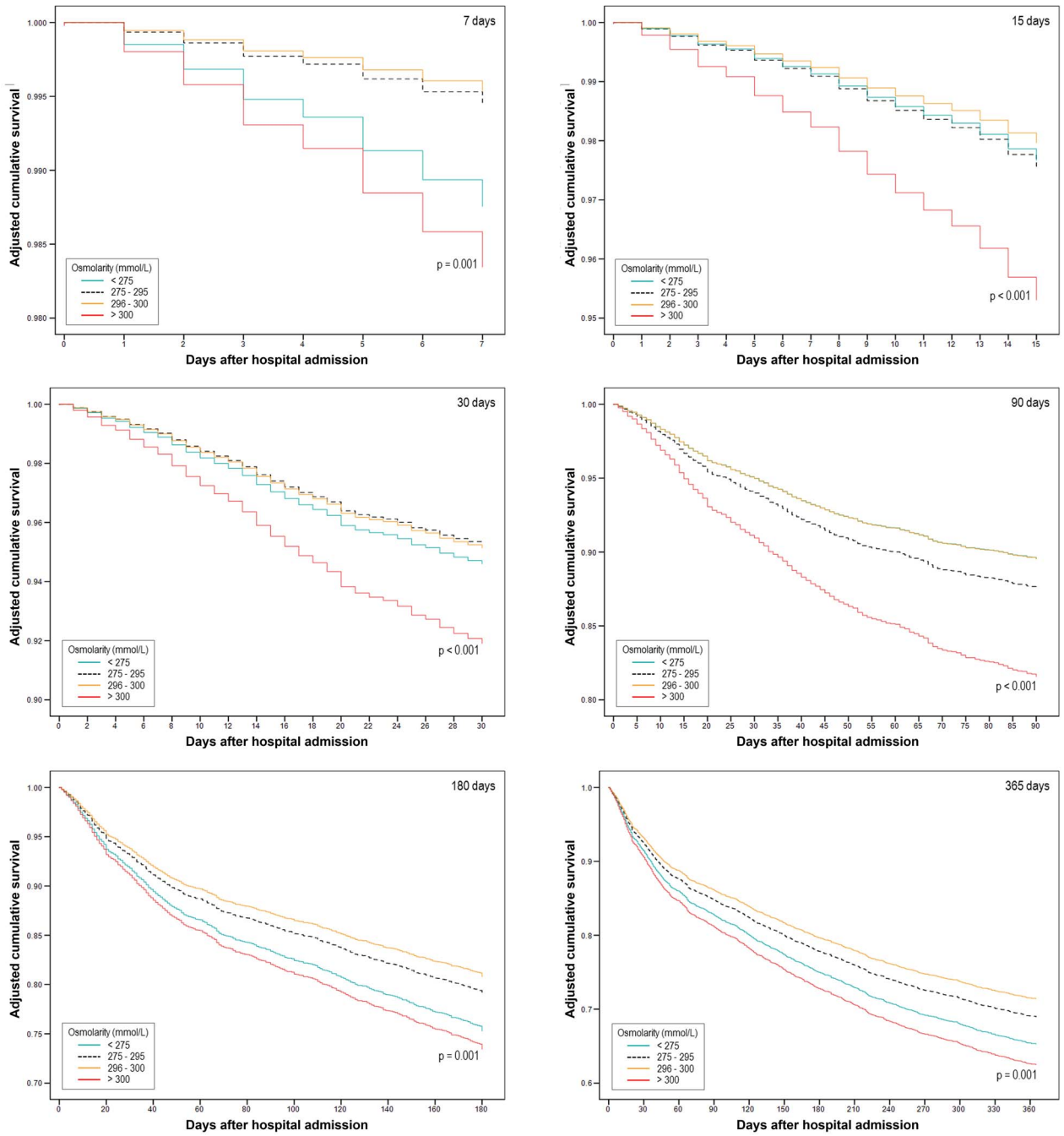

Figure 3. Adjusted survival curves (multivariate Cox regression analysis) at different time interval from hospital admission for patients belonging to different osmolarity categories.

potential prognostic factors in order to identify the most vulnerable patients and to establish appropriate therapeutic interventions.

Supplementary Data: Supplementary data mentioned in the text are available to subscribers in *Age and Ageing* online.

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